
mas-est Guide

Prerequisites · Install · Upgrade · What gets installed · Log in with the demo users · Connection details · Connect S3 to Manage · Connect SMTP · Day-2 operations · Uninstall

When something fails, see Troubleshooting. What the toolkit doesn't do is in Known limitations.

Placeholders: `<instance>` is your MAS instance ID, `<cluster-domain>` the cluster's apps domain, `<workspace>` the MAS workspace ID.

Prerequisites

Need	Detail
Local tools	podman, oc, bash 3.2 or later
Cluster access	oc logged in as cluster-admin. The install creates namespaces, ClusterRoles and OLM resources.
MAS API key	Name and value, with the userAdmin and systemAdmin permissions. Preflight rejects a key without them.
Storage	A block (RBD) storage class. Preflight recommends one.
MAS version	9.0.x or 9.1.x. On MAS 9.0, leave out OIDC. MAS 9.2 has not been tested yet.

Install

1. Bootstrap the CLI

```
export MAS_EST_IMAGE='quay.io/lee_forster/mas-external-services-tool:v0.1.13'
mkdir -p "$HOME/mas-est"
podman run -ti --rm -v "$HOME/mas-est:/tmp" --pull always "$MAS_EST_IMAGE" bootstrap
↪ --force
export PATH="$HOME/mas-est:$PATH"
mas-est version
```

version must print 0.1.13. --force overwrites an earlier runtime and is safe on a first install.

2. Preflight

```
mas-est preflight
```

Enter the MAS SCIM URL, `https://api.<instance>.<cluster-domain>/scim/v2`, and the API key. Success looks like this:

```
[ok] oc: oc CLI found on PATH
[ok] cluster-login: logged in as kube:admin against https://api.<cluster-domain>:6443
[ok] storage-classes: discovered 5 storage classes; recommended=localblock
[warn] storage-selection: default storage class ocs-storagecluster-cephfs looks filesystem-backed; preferring localblock for PostgreSQL
```

```
[ok] mas-base-url: parsed MAS host api.<instance>.<cluster-domain>
[ok] mas-api-key: MAS API key <token-name> authenticated and is authorized for the SCIM API
[ok] mas-route-lookup: matched route mas-<instance>-core/<instance>-api
```

`install` runs two more checks before it starts: whether MAS exposes the OIDC API, and whether login providers named `default` already exist.

3. Install

```
mas-est install
```

Prompt	What to answer
Products to install	LDAP directory, Keycloak identity provider, SCIM bridge, S3 object storage, SMTP Mailpit capture server. SCIM adds Keycloak and LDAP; Keycloak adds LDAP.
MAS SCIM base URL, MAS API token name, MAS API token value	Same as preflight
Workspace ID, MAS profile ID	The workspace is filled in when MAS has one. Profile ID defaults to demo.
MAS instance ID, MAS core namespace	Filled in from the SCIM URL
Configure MAS authentication providers?	Yes. Then choose <code>ldap</code> , <code>oidc</code> , <code>saml</code> . On MAS 9.0 leave out <code>oidc</code> .
Primary storage class, SCIM bridge storage class	The block class preflight recommended

Answers marked `[derived]` were worked out for you. Confirm at `Proceed with install?`.

The install runs as a Job named `mas-est-install` in the `mas-est` namespace and streams its log to your terminal, reconnecting if the connection drops. Ctrl-C detaches from the log. The Job keeps running.

```
mas-est logs --component install-job --follow # reattach
oc delete job mas-est-install -n mas-est      # cancel, or clear a finished Job before
↪ re-running
mas-est install --local                        # run on this machine instead of in the
↪ cluster
```

Success ends with the summary:

```
Install summary (total 17m27s)
  ✓ Install LDAP, Keycloak, SCIM bridge      (5m44s)
  ✓ Install MinIO S3                          (1m47s)
  ✓ Install Mailpit SMTP                     (3s)
  ✓ Configure MAS LDAP / OIDC / SAML         (9m53s)
```

The Job's ServiceAccount gets a ClusterRole close to `cluster-admin`. The install reads secrets in the MAS namespaces and runs `pods/exec` on MongoDB to link the SCIM users. The permissions are applied by the CLI you run, not by the Job image, so re-bootstrap before installing after an upgrade. If your cluster's policy forbids this, use `--local`.

Scripted install, with every value as a flag:

```
mas-est install --non-interactive \
  --components ldap,keycloak,scim,s3,smtp \
  --mas-base-url 'https://api.<instance>.<cluster-domain>/scim/v2' \
  --mas-api-token-name '<name>' --mas-api-token-value '<value>' \
  --workspace-id '<workspace>' --mas-instance-id '<instance>' \
  --configure-mas-auth --mas-auth-providers ldap,oidc,saml \
  --storage-class '<block-class>' --scim-bridge-storage-class '<block-class>'
```

4. Verify

```
mas-est status
```

Every deployment shows ready=1/1, the CSV shows phase=Succeeded, and every PVC shows phase=Bound. Then log in as one of the demo users.

Upgrade

Re-run bootstrap with the new tag. Exporting a new MAS_EST_IMAGE doesn't replace the extracted binary, and install refuses to run while the two differ.

```
export MAS_EST_IMAGE='quay.io/lee_forster/mas-external-services-tool:v<new-version>'
podman run -ti --rm -v "$HOME/mas-est:/tmp" --pull always "$MAS_EST_IMAGE" bootstrap
↪ --force
mas-est version
```

Nothing in the cluster changes until you run `mas-est install` again.

What gets installed

Component	In namespace mas-est	In namespace mas-<instance>-core
MAS IAM operator	CSV mas-iam-operator.v0.0.15, from CatalogSource mas-iam-operator in openshift-marketplace	
OpenLDAP	deployment/mas-est-iam-openldap	idpcfg/<instance>-ldap-default-system
Keycloak, realm maximo, with PostgreSQL	deployment/mas-est-iam, route mas-est-iam	idpcfg/<instance>-oidc-default-system, idpcfg/<instance>-saml-default-system
SCIM bridge	deployment/scim-bridge	SCIM profile demo
MinIO	deployment/mas-minio, routes mas-minio-api and mas-minio-console	objectstoragecfg/ibm-mas-<instance>- objectstoragecfg-system
Mailpit	deployment/mas-mailpit, route mas-mailpit	

Component	In namespace mas-est	In namespace mas-<instance>-core
Self-registration		configmap/<instance>-selfreg. Users are created on first login with PREMIUM entitlement and Manage in <workspace>.

MAS calls a login provider an IDPCfg. The provider ID is always default, which is what makes MAS administration show each provider as configured. If providers with that ID already exist, the install overwrites them. Back them up first:

```
oc get idpcfg -n mas-<instance>-core -o yaml > idpcfg-backup.yaml
```

The install also raises the memory limit of <instance>-entitymgr-idpcfg to 2 Gi through the Suite CR, because the MAS default of 512 Mi runs out while the providers are applied. --idpcfg-memory-limit off skips this.

Users and identity	
Name	Status
LDAP Authentication	Ready
OIDC authentication	Ready
Push Notification	Not configured
SAML Authentication	Ready
SMTP	Not configured
User Registry Synchronization	Not configured

Figure 1: MAS administration, Users and identity, with LDAP, OIDC and SAML authentication all Ready

Log in with the demo users

User	Directory	Log in with	Password
ldap.user1, ldap.user2	OpenLDAP	LDAP	mas-est ldap-info --show-user- passwords
oidc.user1, oidc.user2	Keycloak	OIDC	maxadmin
saml.user1, saml.user2	Keycloak	SAML	maxadmin
scim.user1, scim.user2	Keycloak	OIDC	maxadmin

LDAP, OIDC and SAML users are created in MAS on their first login. SCIM users are created by the SCIM bridge before anyone logs in.

Use a private browser window for every login. Keycloak keeps you signed in after you log out of MAS, so the next login in the same window returns the previous user.

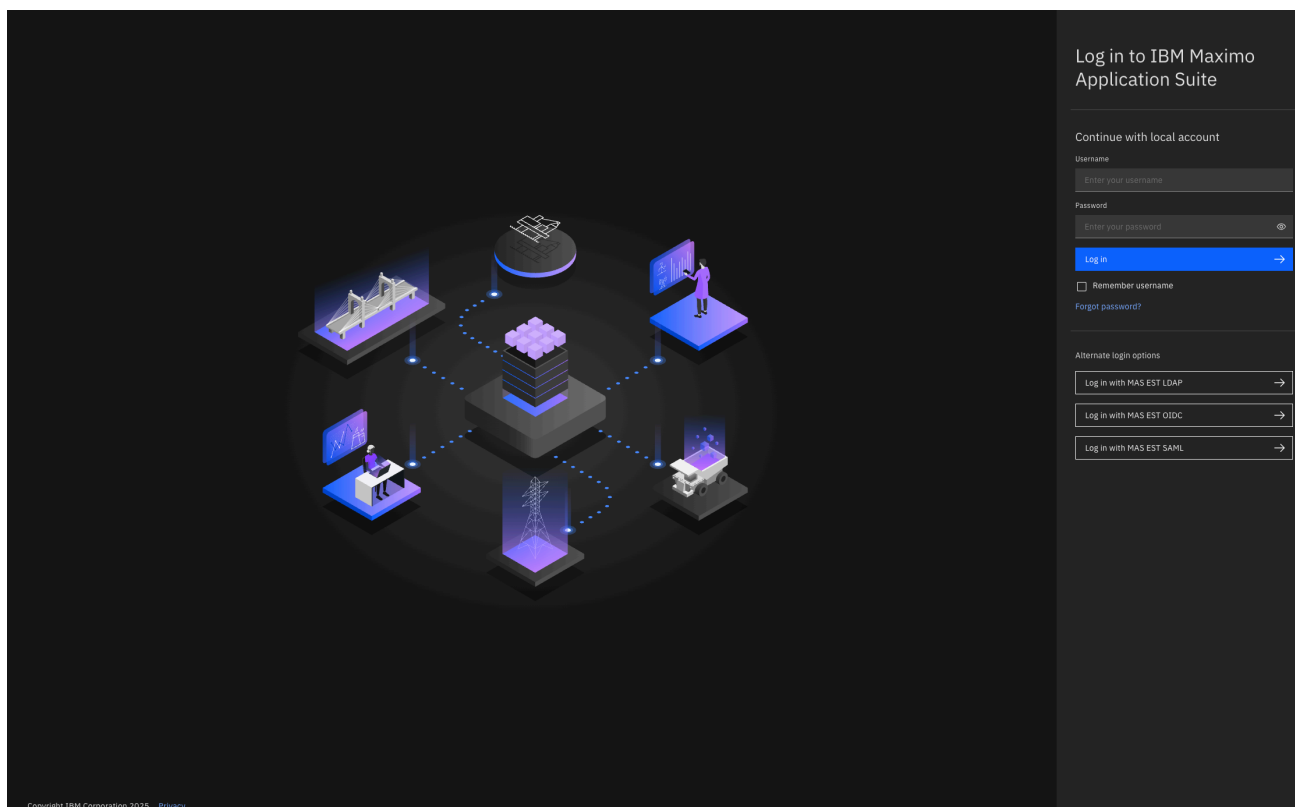


Figure 2: MAS login page with the Log in with MAS EST LDAP, OIDC and SAML options

IBM Maximo Application Suite Users

Users

Manage your suite users, their entitlements, and access rights.

View: Default view (shared) Search

Clear all filters

Create user

User ID	Person	Primary email	Access type	Entitlement	Status	Sync status	Account Lock
LDAP.USER1	LDAP.USER1	ldap.user1@demo.local	Concurrent	Self Service	Newly self-registered	Complete	
LDAP.USER2	LDAP.USER2	ldap.user2@demo.local	Concurrent	Self Service	Newly self-registered	Complete	
OIDC.USER1	OIDC.USER1	oidc.user1@demo.local	Concurrent	Self Service	Newly self-registered	Complete	
OIDC.USER2	OIDC.USER2	oidc.user2@demo.local	Concurrent	Self Service	Newly self-registered	Complete	
SAML.USER1	SAML.USER1	saml.user1@demo.lo...	Concurrent	Self Service	Newly self-registered	Complete	
SAML.USER2	SAML.USER2	saml.user2@demo.lo...	Concurrent	Self Service	Newly self-registered	Complete	
SCIM.USER1	SCIM.USER1	scim.user1@demo.local	Concurrent	Self Service	Active	Complete	
SCIM.USER2	SCIM.USER2	scim.user2@demo.local	Concurrent	Self Service	Active	Complete	

Items per page: 10 1-8 of 8 items 1 1 of 1 pages

Figure 3: MAS Users after every demo user has logged in once: SCIM users Active, the others Newly self-registered

Add a SCIM user

1. Open the Keycloak admin console at the `mas-est-iam` route, realm `maximo`. The admin password is in secret `mas-est-iam-bootstrap-admin`.
2. Create a user whose username starts with `scim.`, give it an email address, and add it to the group `mas-scim-users`.

- Wait up to 5 minutes. `mas-est logs` shows `action=create username=scim.<name>` when the bridge has created it in MAS.
- Link the user to its OIDC identity. Without this step it can't log in or reach Manage.

```
mas-est mas-auth apply --mas-instance-id <instance> --self-reg-workspace <workspace>
```

On MAS 9.0 add `--providers ldap,saml`.

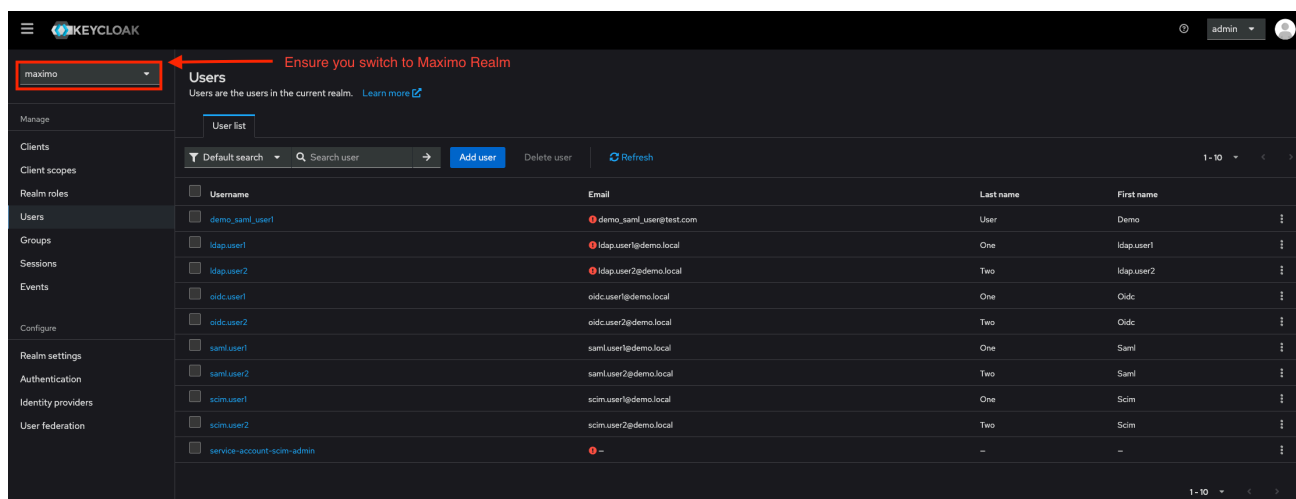


Figure 4: Keycloak admin console, realm maximo, showing the demo users

Connection details

```
mas-est details          # --component ldap|oidc|saml|s3|smtp; --show-secrets
mas-est ldap-info        # --show-password; --show-user-passwords
```

Both print what MAS is configured with. The raw values are in these resources in `mas-est`:

Resource	Keys
<code>secret/mas-est-ldap-connection</code>	<code>url</code> , <code>baseDN</code> , <code>bindDN</code> , <code>bindPassword</code> , <code>userIdMap</code> , and <code>ca.crt</code> when the LDAP CA was found
<code>secret/mas-est-oidc-connection</code>	<code>issuerUrl</code> , <code>discoveryUrl</code> , <code>authorizationEndpoint</code> , <code>tokenEndpoint</code> , <code>jwtEndpoint</code> , <code>clientId</code> , <code>clientSecret</code> , <code>realm</code> , <code>redirectUri</code>
<code>secret/mas-est-saml-connection</code>	<code>entityId</code> , <code>acsUrl</code> , <code>sloUrl</code> , <code>idpMetadataUrl</code> , <code>nameIdFormat</code> , and <code>idpMetadata</code> when the metadata fetch succeeded
<code>secret/mas-est-s3-connection</code>	<code>provider</code> , <code>endpoint</code> , <code>manageEndpoint</code> , <code>externalEndpoint</code> , <code>consoleUrl</code> , <code>accessKey</code> , <code>secretKey</code> , <code>region</code> , <code>bucket</code> , <code>siblingBuckets</code>
<code>configmap/mas-est-smtp-connection</code>	<code>host</code> , <code>port</code> , <code>from</code> , <code>webUI</code> , <code>tls</code> , <code>authentication</code> , <code>relayEnabled</code>

```
oc get secret mas-est-oidc-connection -n mas-est -o jsonpath='{.data.clientSecret}' |
  ↪ base64 -d
```

LDAP settings

Setting	Value
URL, inside the cluster	ldaps://mas-est-iam-openldap.mas-est.svc.cluster.local:636
Bind DN	cn=admin,dc=demo,dc=local. Password: secret mas-est-iam-openldap-admin, key password
Base DN	dc=demo,dc=local
Users, groups	ou=users,dc=demo,dc=local, ou=groups,dc=demo,dc=local
User attribute	uid
Group object class, member attribute	groupOfUniqueNames, uniqueMember

From your machine: `oc -n mas-est port-forward svc/mas-est-iam-openldap 1636:636`.

Connect S3 to Manage

The install connects MinIO to MAS Suite through the `ObjectStorageCfg`. Manage attachments, called doclinks, need these steps.

1. In Manage, open **System Properties** and set:

Property	Value
<code>mxe.cosendpointuri</code>	<code>http://mas-est.svc.cluster.local:9000</code>
<code>mxe.cosbucketname</code>	<code>mas-s3-demo</code>
<code>mxe.cosregion</code>	<code>us-east-1</code> . Add it with New Row if it doesn't exist.
<code>mxe.cosaccesskey</code>	<code>minioadmin</code>
<code>mxe.cossecretkey</code>	<code>Secretmas-minio-root, key MINIO_ROOT_PASSWORD</code>
<code>mxe.attachmentstorage</code>	The class name in the block below
<code>mxe.doclink.doctypes.defpath</code>	<code>cos:doclinks</code>
<code>mxe.doclink.doctypes.topLevelPaths</code>	<code>cos:doclinks</code>
<code>mxe.doclink.path01</code>	<code>cos:doclinks=<manage-ui-base-url></code>
<code>mxe.doclink.securedAttachment</code>	<code>True</code>

Value for `mxe.attachmentstorage`:

```
com.ibm.tivoli.maximo.oslc.provider.COSAttachmentStorage
```

2. Click **Live Refresh**.
3. In **Document Types**, set each document type's default path to `cos:doclinks/<name>`, for example `cos:doclinks/attachment`.

4. Restart Manage:

```
oc rollout restart deployment/<instance>-<workspace>-all -n mas-<instance>-manage
```

Use the endpoint above, not the MinIO route. Manage addresses buckets as <bucket> . <host> . The install creates a Service for each bucket name that resolves only inside the cluster, and the route returns HTTP 503 or certificate errors. Without `mx.cosregion`, every request fails with `SignatureDoesNotMatch`.

Browse the buckets at the `mas-minio-console` route as user `minioadmin`, with the `MINIO_ROOT_PASSWORD` key of secret `mas-minio-root`.

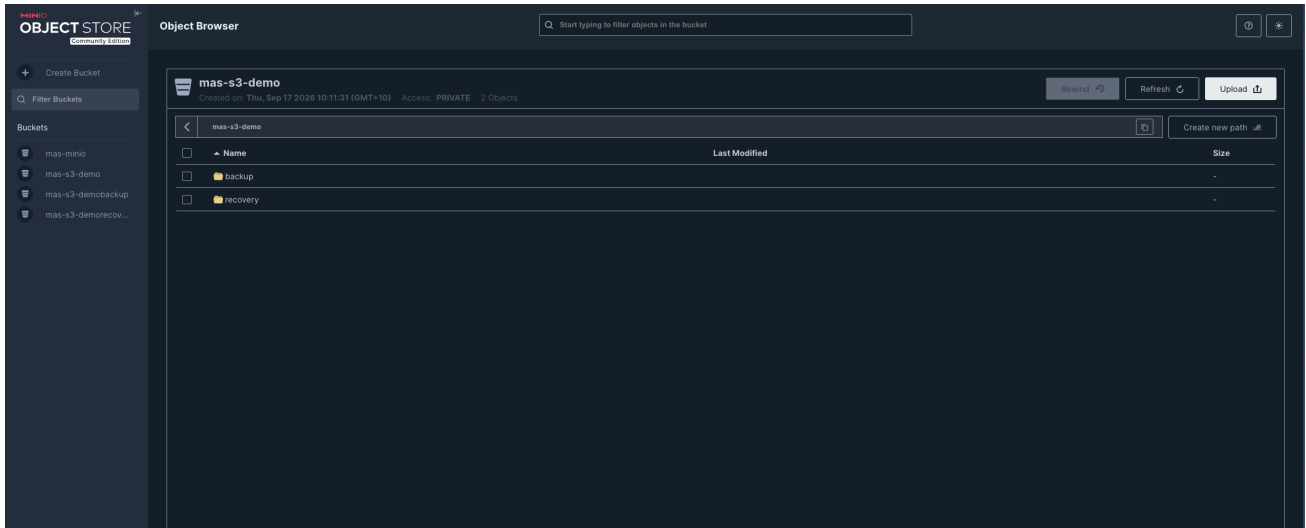


Figure 5: MinIO console showing bucket `mas-s3-demo` with its backup and recovery folders, and the sibling buckets in the sidebar

Connect SMTP

Mailpit captures every message and shows it at the `mas-mailpit` route. It delivers nothing unless you add a relay. The install doesn't point MAS at it. Set these in MAS administration:

Setting	Value
Host	<code>mas-mailpit.mas-est.svc.cluster.local</code>
Port	1025
TLS	Off
Authentication	None

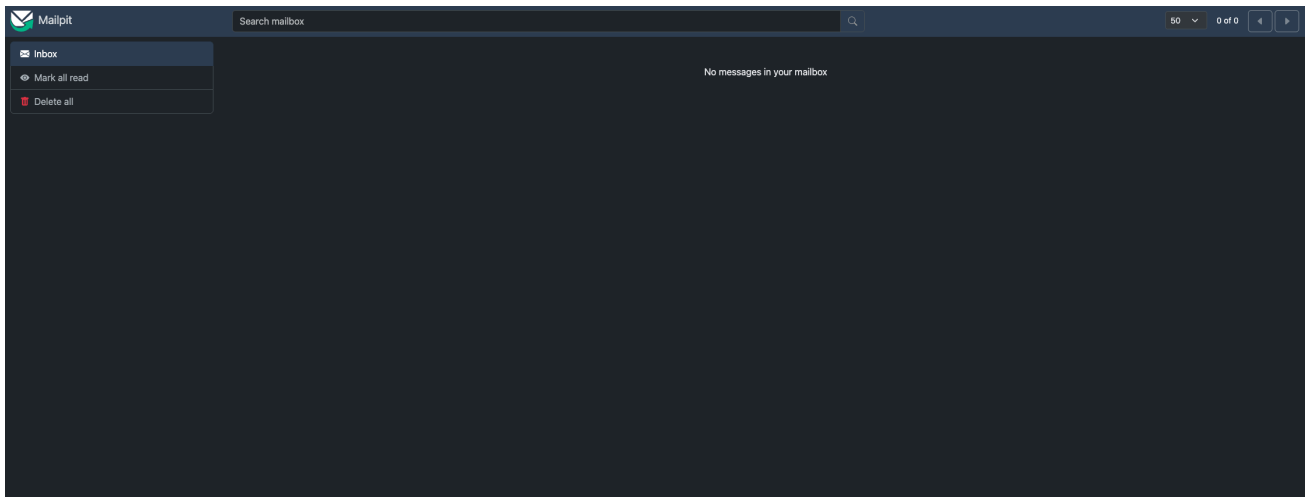


Figure 6: Mailpit inbox at the mas-mailpit route

To forward captured mail to real inboxes, install with relay flags. They also work with `mas-est smtp install-mailpit`.

```
--smtp-relay-host smtp.gmail.com --smtp-relay-port 587 --smtp-relay-starttls \  
--smtp-relay-username 'lab@example.com' --smtp-relay-password "$APP_PASSWORD" \  
--smtp-relay-from 'lab@example.com'
```

The MAS settings stay the same. Gmail needs an App Password. For SendGrid the username is the literal apikey. Microsoft 365 needs OAuth, which Mailpit doesn't support.

Day-2 operations

Health and logs

```
mas-est status  
mas-est logs --component bridge --tail 300 # --follow to stream
```

Log components: operator, keycloak, openldap, bridge, profile-bootstrap, minio, minio-init, smtp, install-job.

SCIM bridge settings

The bridge reads `configmap/scim-bridge-config` and `secret/scim-bridge-secret` at startup. These commands restart it for you:

```
mas-est config view  
mas-est config set mas-api-token --token-name '<name>' --token-value '<value>' # rotate  
↪ the MAS API key  
mas-est config set bridge --log-level debug # or  
↪ --payload-logging true  
mas-est restart bridge # after  
↪ editing the ConfigMap by hand
```

The bridge syncs a user only if it is in the Keycloak group named by SCIM_BRIDGE_INCLUDE_GROUPS (mas-scim-users) and its username starts with SCIM_BRIDGE_INCLUDE_USERNAME_PREFIX (scim.). The full setting list is in services/scim-bridge/README.md.

Payload logging writes usernames, names and email addresses to the log. Turn it off when you're done.

Re-apply or remove the MAS login providers

```
mas-est mas-auth apply --mas-instance-id <instance> --self-reg-workspace <workspace> #  
↪ re-create providers and re-link SCIM users  
mas-est mas-auth delete --mas-instance-id <instance> #  
↪ remove all three; --providers to pick
```

apply reads the API key from secret/scim-bridge-secret. Always pass --self-reg-workspace, or self-registration falls back to a workspace named workspace.

Report a problem

```
mas-est support-bundle
```

Writes a mas-est-support-mas-est-<timestamp> directory with status, events, pod and route listings, ConfigMaps and component logs. No Secret values are written. Check it for customer hostnames before sharing it outside the team.

Uninstall

```
mas-est uninstall # also deletes SCIM profile demo  
mas-est uninstall --skip-profile-delete # keeps the profile
```

Deletes the mas-est namespace. It leaves behind:

- The login providers. Remove them with `mas-est mas-auth delete --mas-instance-id <instance>`.
- `configmap/<instance>-selfreg` and the Suite CR memory bump.
- MAS user records in MongoDB and Manage.
- `clusterrole/mas-est-installer`, `clusterrolebinding/mas-est-installer-mas-est`, the SCC `mas-est-iam-openldap-tls-generator`, and the CRD `masiamstacks.iam.mas.ibm.com`.
- CatalogSource `mas-iam-operator` in openshift-marketplace.

Reinstalling onto the same MAS with leftover user records can break self-registration and Manage sync. The fixes are in Troubleshooting.